Euclid's Elements — Book I
Proposition 19

CGJteamLab

In any triangle the greater angle is subtended by the greater side.

1 Introduction

Euclid's Proposition I.19 is the converse of Proposition I.18. In a nondegenerate triangle, the greater angle is opposite the greater side.

For a triangle ABC, if

$$\angle ACB < \angle ABC,$$

then

$$AB < AC.$$

The classical proof is indirect. One compares the two opposite sides AB and AC. Exactly one of the following three possibilities must occur:

$$AB \cong AC, \qquad AB < AC, \qquad AC < AB.$$

The first case contradicts Proposition I.5, because equal sides would imply congruent base angles. The third case contradicts Proposition I.18, because the greater side AB would force the opposite angle at C to be greater than the angle at B. Hence only

$$AB < AC$$

remains.

The formal reconstruction makes this trichotomy explicit through the new general theorem

`hilbert_segment_trichotomy`.

It also isolates a general irreflexivity theorem for strict angle comparison:

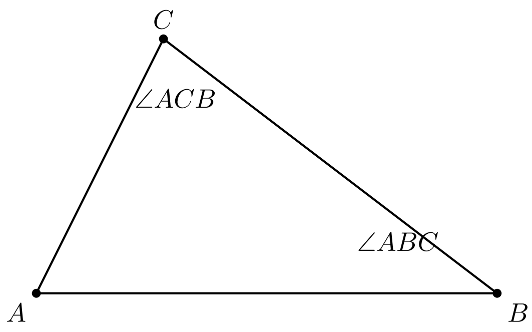
`hilbert_angleLess_irrefl`.

These two results are reusable structural components of the Hilbert layer rather than local devices specific to Proposition I.19.

2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$\angle ACB < \angle ABC.$$



The theorem asserts that the side opposite the larger angle must be larger:

$$AB < AC.$$

No new auxiliary point is required. The proof is entirely comparative: it uses the trichotomy of the two sides and eliminates the two alternatives incompatible with the assumed angle inequality.

3 Classical Proof

Assume

$$\angle ACB < \angle ABC.$$

Compare the two sides AB and AC.

If

$$AB = AC,$$

then triangle ABC is isosceles. Proposition I.5 gives

$$\angle ABC = \angle ACB,$$

contradicting the hypothesis that

$$\angle ACB < \angle ABC.$$

If instead

$$AC < AB,$$

then Proposition I.18, applied to triangle ACB, yields

$$\angle ABC < \angle ACB,$$

which again contradicts the original hypothesis.

The only remaining possibility is

$$AB < AC.$$

Thus the greater angle is subtended by the greater side.

The classical dependency pattern is therefore

$$\begin{array}{ccc} & \angle ACB < \angle ABC & \\ & \downarrow & \\ & AB \cong AC \ \vee \ AB < AC \ \vee \ AC < AB & \\ & \downarrow & \\ AB \cong AC & AB < AC & AC < AB \\ \downarrow & & \downarrow \\ \text{I.5 contradiction} & \text{desired case} & \text{I.18 contradiction} \end{array}$$

4 Segment Trichotomy

The key structural theorem introduced for the formal proof is

`hilbert_segment_trichotomy`.

Its conclusion has the form

$$AB \cong CD \ \vee \ AB < CD \ \vee \ CD < AB.$$

The proof does not introduce a new axiom. It is derived from segment construction and the order theory of collinear points.

A copy of AB is laid off on the ray CD, producing a point X such that

$$CX \cong AB.$$

If $X = D$, then

$$AB \cong CD.$$

Otherwise, C, D, X are distinct collinear points. The Hilbert betweenness trichotomy gives three possibilities. The case

$$D - C - X$$

is incompatible with the fact that X lies on the ray CD. The remaining two orders give respectively

$$CD < CX$$

and

$$CX < CD.$$

Transporting these inequalities through

$$CX \cong AB$$

yields the required segment trichotomy.

Thus the theorem packages a general completeness property of strict segment comparison.

5 Irreflexivity of Strict Angle Comparison

The second structural theorem used in I.19 is

`hilbert_angleLess_irrefl`.

It states

$$\neg(\alpha < \alpha).$$

In the synthetic definition of `HilbertAngleLess`, the relation

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

Therefore

$$\alpha < \alpha$$

would imply that an angle is congruent to one of its own proper interior subangles. This is excluded by the previously proved theorem

`hilbert_interior_subangle_not_congruent_whole`.

The irreflexivity result provides exactly the contradiction mechanism needed in both non-target branches of Proposition I.19.

6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three stages.

6.1 Nondegeneracy of the Compared Side

From the noncollinearity hypothesis

$$\neg\,\mathrm{Collinear}(A, B, C)$$

the proof first derives

$$A \neq C.$$

This is the nondegeneracy condition required to apply segment construction in `hilbert_segment_trichotomy`.

6.2 Apply Segment Trichotomy

The theorem

`hilbert_segment_trichotomy`

is applied to the segments AB and AC.

This produces three branches:

$$AB \cong AC, \qquad AB < AC, \qquad AC < AB.$$

The middle branch is exactly the desired conclusion.

6.3 Eliminate the Two Impossible Branches

In the congruent-side branch, Proposition I.5 gives

$$\angle ABC \cong \angle ACB.$$

The assumed strict inequality

$$\angle ACB < \angle ABC$$

is transported across that congruence, producing

$$\angle ACB < \angle ACB.$$

This contradicts `hilbert_angleLess_irrefl`.

In the branch

$$AC < AB,$$

Proposition I.18 applied to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Combining this with the original inequality by transitivity yields

$$\angle ACB < \angle ACB,$$

again contradicting irreflexivity.

Only

$$AB < AC$$

remains.

7 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_19
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hAngle :
      HilbertAngleLess Geo A C B A B C) :
    HilbertSegmentLess Geo A B A C := by
```

The proof first derives the nondegeneracy of AC:

```
  have hAC : A != C :=
    hilbert_noncollinear_ne_first
      Geo A C B
      (by
        intro h
        exact hABC
          (PrimCollinearRotate Geo A C B h))
```

Then segment trichotomy is applied:

```
rcases
    hilbert_segment_trichotomy
      Geo
      A B
      A C
      hAC with
  hEq | hABltAC | hACltAB
```

The middle branch closes immediately:

```
$\cdot$ exact hABltAC
```

For the congruent-side branch, Proposition I.5 gives:

```
have hIso :
    Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
    Geo
    A B C
    hABC
    hEq
```

The strict inequality is transported to the congruent target angle:

```
have hSame :
    HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_transport_right
    Geo
    A C B
    A B C
    A C B
    hAngle
    hACB
    hIso
```

This contradicts irreflexivity:

```
exact
  False.elim
    ((hilbert_angleLess_irrefl
        Geo A C B)
      hSame)
```

In the final branch, Proposition I.18 gives the opposite strict inequality:

```
have hOpp :
    HilbertAngleLess Geo A B C A C B :=
  euclid_proposition_18
    Geo
    A C B
    hACB
    hACltAB
```

Transitivity then creates an impossible strict self-comparison:

```
have hCycle :
```

```
    HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_trans
    Geo
    A C B
    A B C
    A C B
    hAngle
    hOpp
```

and finally:

```
exact
  False.elim
    ((hilbert_angleLess_irrefl
        Geo A C B)
      hCycle)
```

8 Relationship with Earlier Propositions

8.1 Proposition I.5

I.5 is used to eliminate the equality case. If

$$AB \cong AC,$$

then the base angles are congruent:

$$\angle ABC \cong \angle ACB.$$

This is incompatible with the assumed strict angle inequality.

8.2 Proposition I.18

I.18 is used to eliminate the reversed side inequality. If

$$AC < AB,$$

then applying I.18 to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Together with the hypothesis

$$\angle ACB < \angle ABC,$$

this produces a strict cycle and therefore a contradiction.

8.3 Logical Relation between I.18 and I.19

The two propositions form a converse pair:

$$AB < AC \quad \Longrightarrow \quad \angle ACB < \angle ABC$$

for I.18, and

$$\angle ACB < \angle ABC \quad \Longrightarrow \quad AB < AC$$

for I.19.

Together they establish the equivalence between ordering of opposite sides and ordering of opposite angles in a nondegenerate triangle.

9 Hilbert and Book Zero Components Used

9.1 `hilbert_segment_trichotomy`

This theorem provides the complete three-way comparison of two segments and is the main new structural component required by I.19.

9.2 `hilbert_angleLess_irrefl`

This theorem excludes strict self-comparison of an angle. It is used in both contradictory branches of the proof.

9.3 `hilbert_isosceles_base_angles`

This is the formal version of Proposition I.5 needed to eliminate the equality case.

9.4 `hilbert_angleLess_transport_right`

This theorem transports a strict angle comparison across congruence of the containing angle. It converts the equality branch into an impossible self-comparison.

9.5 `hilbert_angleLess_trans`

Transitivity combines the original strict inequality with the opposite inequality supplied by I.18.

10 Formalization Notes

Proposition I.19 is formally much shorter than Proposition I.18, but its proof is logically revealing.

The argument does not construct a new geometric configuration. Instead, it performs a case analysis on the two sides whose order is to be determined. Segment trichotomy gives three possibilities. Two of them are incompatible with the assumed strict angle inequality.

The equality branch is eliminated by the isosceles-triangle theorem: equal sides force the corresponding angles to be congruent, contradicting strict angle comparison.

The reverse-inequality branch is eliminated by Proposition I.18. If the supposedly smaller side were actually larger, I.18 would force the opposite angle inequality. Together with the hypothesis this would produce a strict angle inequality from an angle to itself, contradicting irreflexivity.

Thus the proof is a characteristic formal converse argument:

$$\text{trichotomy} \ + \ \text{direct theorem I.18} \ + \ \text{irreflexivity} \quad \Longrightarrow \quad \text{I.19.}$$

This organization is useful beyond Proposition I.19. It shows that converse comparison theorems need not be proved by reproducing the geometric construction of the direct theorem. Once strict order has trichotomy and irreflexivity, the converse can often be obtained by eliminating the impossible alternatives.

11 Dependency Summary

The formal dependency structure is

$$
\begin{array}{c}
\text{Hilbert incidence + order + congruence}\\
\downarrow\\
\texttt{hilbert\_segment\_trichotomy} \qquad \texttt{hilbert\_angleLess\_irrefl}\\
\downarrow\\
AB \cong AC \ \vee \ AB < AC \ \vee \ AC < AB\\
\downarrow\\
\begin{array}{ccc} \text{I.5} & & \text{I.18}\\ \downarrow & & \downarrow\\ \text{contradiction} & AB < AC & \text{contradiction}\end{array}\\
\downarrow\\
\texttt{euclid\_proposition\_19}.
\end{array}
$$

No new geometric axiom is introduced.

12 Conclusion

Proposition I.19 is formally shorter than I.18, but it exposes an important structural issue: the proof requires a complete comparison principle for segments.

Once the theorem `hilbert_segment_trichotomy` is available, the Euclidean argument becomes exactly the expected three-case proof. Equal sides are ruled out by I.5, the reversed inequality is ruled out by I.18, and the remaining case is the desired conclusion.

The proof also clarifies the role of strict angle order. Instead of introducing an explicit asymmetry theorem, the development uses the more fundamental statement `hilbert_angleLess_irrefl` together with transitivity. This keeps the order theory modular and avoids unnecessary duplication.

Thus I.19 strengthens the reusable Hilbert layer in two directions: completeness of segment comparison and irreflexivity of strict angle comparison. The Euclidean theorem itself then becomes a concise application of those general principles.

The completed Lean development compiles cleanly and contains no `sorry`.